Heart Disease Prediction using PyTorch Neural Network – 5 PAGE REPORT

Lay Introduction: Heart disease remains one of the leading causes of death worldwide, making early detection vital for prevention and treatment. This project uses data and artificial intelligence to help predict whether a person is likely to have heart disease based on clinical measurements. Using patient data from the well-known Cleveland Heart Disease dataset, I trained a computer model to recognize patterns linked to heart disease. The dataset includes information such as age, blood pressure, cholesterol, and test results like ECG readings and exercise responses that doctors use and collect. By learning from these factors, the model can estimate a person's risk of heart disease and even assign a "risk score" showing how likely the condition is. The model achieved high accuracy (about 88%) and identified key predictors like blocked arteries, chest pain type, and maximum heart rate. While the dataset is small and mostly male, the results show how machine learning can help support decision making in a medical setting. Implications of this work include a tool for use by clinicians as an assistant, the demonstration of techniques that prevent models from memorizing data and instead focus on learning patterns, and improving identification of at-risk patients using data collected in the past.

Project Overview

1. Background: Cardiovascular disease is a leading cause of mortality worldwide, particularly in the United States. Early and accurate detection is crucial for effective intervention and treatment. According to Cleveland Clinic, heart disease, a subset of cardiovascular diseases, involves issues such as narrowing of blood vessels, congenital heart and vessel problems, and arrhythmias. These conditions can collectively stress the cardiovascular system and lead to heart failure, increased risk of heart attacks, and chronic pain. High blood pressure, lack of physical activity, genetics, and cholesterol levels can contribute to the development of heart disease. Clinical decision-making can be enhanced by the adoption of data-driven, AI-integrated solutions that take advantage of ample amounts of patient data. This project specifically harnesses the Cleveland Heart Disease dataset from the UCI Machine Learning Repository, a well-known benchmark dataset in medical informatics, which contains 13 clinical features from 303 patients, of which 297 had complete datasets. These features are listed below, and include demographic history, medical history, and diagnostic test results, such as visualizations of major blood vessels. In the future, this simulation of machine learning in cardiovascular diagnostics aims to become a reality.

2. Motivation: The primary motivation is to demonstrate the application of deep learning in a pressing, high-impact region of medicine, cardiovascular disease. While many studies use traditional models like Logistic Regression for this task, this project explores whether a Neural Network can offer superior performance. I also wanted to test whether a neural network with early stopping criteria could avoid overfitting and work on simply 300 or so data points. Furthermore, by conducting a feature importance analysis and statistical analysis, this project aims to showcase the features most important for strong and robust diagnoses of heart disease, including risk factors.

3. Summary

Note on Project Version: This project has a 3-5 page condensed version of the heart disease prediction project for the submission criteria, and a full version with the full analysis. Unlike the full implementation, the 3-5 page version utilizes a simplified training approach with a fixed 30-epoch run on training data without early stopping or validation set separation. The dataset is split 80-20 between training and testing without an intermediate validation partition. Key analyses included in this version are: categorical feature visualizations (bar plots) – corresponding to Figure 2 in the full version – neural network training curves (loss and accuracy), and final model evaluation metrics (confusion matrix, ROC curve, and precision-recall curve), corresponding to Figure 4 from the full version. The model architecture remains consistent with literature-inspired design featuring batch normalization and dropout regularization, but training is conducted to a predetermined epoch count rather than using early stopping criteria. Furthermore, because of the 3-5 page space constraint for the writeup as well, this writeup will be 5 pages long but will only contain figures from the condensed code version (only 2 out of 7 figures). For a full figure list from the expanded version, refer to the IPYNB file.

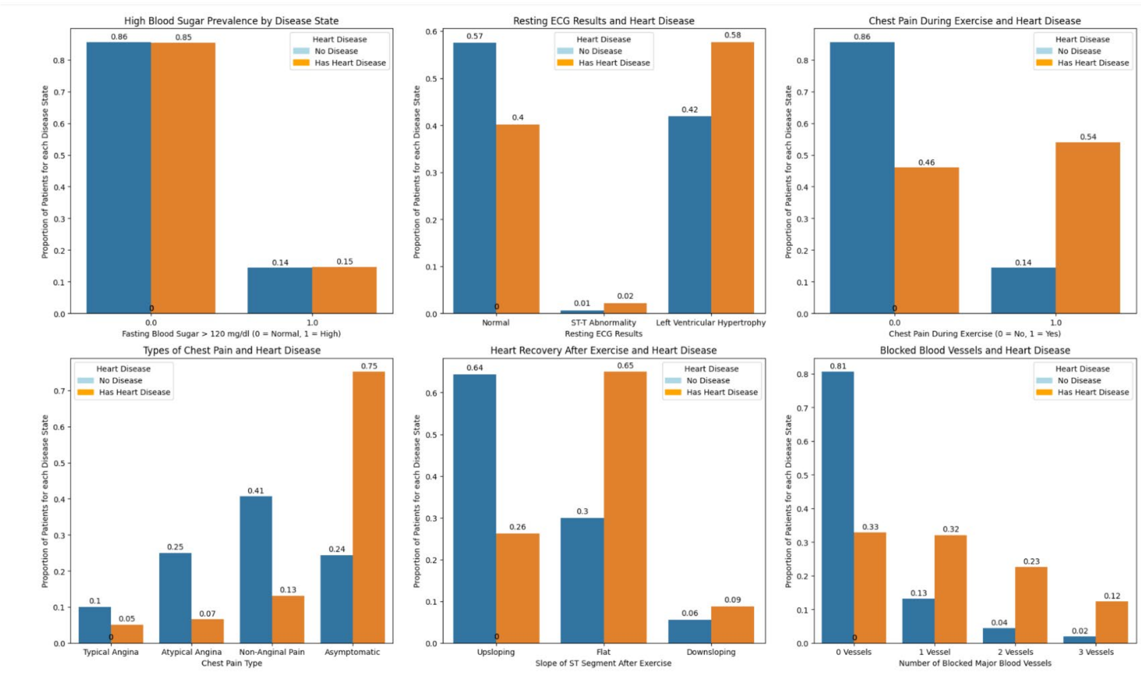
Data Processing: The project begins by loading the Cleveland dataset, handling missing values in the 'ca' and 'thal' columns by removing affected rows, resulting in 297 clean samples. The target variable, which originally indicated disease severity (0-4), was binarized (0 for no disease, 1 for disease). The data was then split into training (65%), validation (15%), and test (20%) sets, with stratification to preserve the target distribution. All features were standardized using StandardScaler as was done in previous assignments for the class, and allows for efficient model training.

Neural Network: A custom PyTorch Neural Network was implemented with two hidden layers (32 and 16 neurons, although running the model was done with 64 and 32 neurons with further testing as this improved accuracy considerably), Batch Normalization, Dropout for regularization, and a Sigmoid output layer. The model was trained with early stopping adapted from Keras to prevent overfitting. The model was trained each epoch on batches of training data and validated on batches of validation data to identify improvements in validation loss/accuracy – if a minimum improvement is not attained, the model training was halted at the given epoch (which in practice occurred between 21-25 epochs). The model was then tested on the held-out testing dataset and performance was analyzed.

Figures

- **Fig 1a-1h (Exploratory Data Analysis):** This series of plots provides a first-part analysis of the data.
    - **Fig 1a:** Pie chart showing distribution of heart disease prevalence in the patient sample, showing a relatively balanced dataset (54% No Disease, 46% Disease).
    - **Fig 1b:** Distribution of patient population in bins of 10, showing that the most common age bin is 55-65 years old. Very few patients are under 35, expected for a disease with a high correlation with aging.
    - **Fig 1c:** Proportion of patients in each age bin that have heart disease, showing that 64% of the patient population sampled between 55-65 years old has heart disease.
    - **Fig 1d, 1e, 1g:** Boxplots show the distribution of cholesterol levels, maximum heart rate, resting blood pressure, and ST depression compared between patients with heart disease and patients without. Although significance testing was not done here, patients without heart disease typically attain higher maximum heart rates, but have lower ST depression values, falling in line with documentation. Cholesterol distribution and Resting BP are roughly aligned between the two disease state groups.
    - **Fig 1h:** Pie chart with sex distribution of patient population showing a significant overrepresentation of male patients (68%).
- **Fig 2a-2f (Categorical Feature Analysis):** These bar plots illustrate the relationship between categorical features and disease prevalence.
    - **Fig 2a:** High blood sugar prevalence (0 for normal blood sugar, 1 for high) compared between patients with and without heart disease, showing similar distributions.
    - **Fig 2b:** Resting ECG results compared between disease state groups. Patients without heart disease were more likely to have normal ECG results, while patients with heart disease had higher rates of left ventricular hypertrophy.
    - **Fig 2c:** Chest pain during exercise – patients without heart disease had significantly higher rates of a "yes", or 1.0, for this feature.
    - **Fig 2d:** Types of chest pain compared between disease state groups. Patients without heart disease surprisingly had higher rates of anginal and non-anginal pain, while patients with heart disease were more likely to be asymptomatic.
    - **Fig 2e:** Slope of ST segment after exercise, indicative of heart recover-ability, showing patients with heart disease having more flat/downsloping ST segments post-exertion, with controls having more upsloping.

- **Fig 2f:** Number of blocked/stenosis-implicated major blood vessels between disease state groups, showing patients with heart disease having significantly higher rates of 1, 2, or 3 major blocked, visualized vessels. Most controls had no vessels blocked.

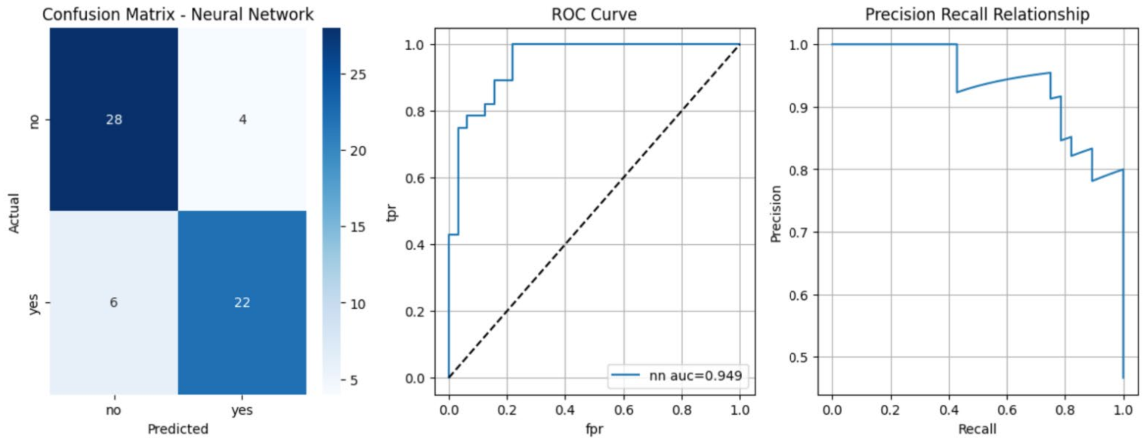


Above: Figures 2a through 2f (top left to bottom right)

Statistical Testing: T-tests with Bonferroni correction identified age - Age, sex - Sex, cp - Chest Pain Type, thalach - Maximum Heart Rate Achieved, exang - Exercise-Induced Angina, oldpeak - ST Depression Induced by Exercise, slope - Slope of Peak Exercise ST Segment, ca - Number of Major Vessels Colored by Fluoroscopy, and thal - Thallium Stress Test Result, as statistically significant predictors of heart disease. The mean, median, standard deviation, and other statistical metrics were shown for these variables/predictors.

- **Neural Network Model:** A custom PyTorch Neural Network was implemented with two hidden layers (64 and 32 neurons), Batch Normalization, Dropout for regularization, and a Sigmoid output layer. The model was trained with early stopping to prevent overfitting.
- **Fig 3a & 3b (Training Curves):** These plots show the model's learning process. Training and validation loss decreased steadily, and accuracy increased, with early stopping triggered at epoch 24 due to early-stopping criteria/regularization.
- **Model Evaluation (Fig 4a-4c):** On the test set, the Neural Network achieved strong performance, specifically an accuracy of 88.3% and AUC-ROC of 0.955, corroborated by figures 4b and 4c showing ROC curve and precision-recall curve. Confusion matrices (4a) show effective classification of both heart disease positive and negative groups, with slightly better performance predicting patients without heart disease.

	precision	recall	f1-score	support
No Heart Disease	0.82	0.88	0.85	32
Disease	0.85	0.79	0.81	28
accuracy			0.83	60
macro avg	0.83	0.83	0.83	60
weighted avg	0.83	0.83	0.83	60



Above: Figures 4A through 4C (right to left).

- **Feature Importance (Fig 5):** Using a permutation-based method, the top 5 most important features were identified as: number of major vessels visualized via fluorophores (ca), chest pain type (cp), thalassemia (thal), maximum heart rate achieved (thalach), and sex (sex). Positive values mean that when randomizing the value of these features, the accuracy was reduced, on average, over four trials per feature. The following tables showcase these top features, the average accuracy drop, and features reducing accuracy – in this case, the only variable for whom randomization increased model accuracy was fasting blood sugar exceeding 120 mg/dl, which fits with the lack of any noticeable difference in the distribution of this feature between the heart disease positive and negative groups.

- **Model Comparison (Fig 6a-6f):** The Neural Network was compared against Logistic Regression, Random Forest, and SVM. It achieved the best scores in Accuracy, Precision, Recall, and F1-score, while SVM had a marginally better AUC-ROC (0.959). This demonstrates the neural networks strong performance despite having a limited dataset, particularly with the help of early stopping criteria and a relatively small-scale structure with reduced neurons per layer. SVMs are also highly usable in this situation due to the smaller dataset, binary classification need, and relatively low number of features used for classification.

- **Risk Stratification (Fig 7a & 7b):** Patients were categorized into Low, Medium, and High-risk groups based on predicted probability, but only for the test dataset with n = 60. Nearly 50% of the patients were at low-risk, while a third were high risk. The probability criteria were set to 0.3 and 0.7 for the three-category-differentiation to reflect the benefit of being safer and taking greater precaution. The distribution of probability values by risk category is also shown in Figure 7b. Finally, the output below the figures showcases the distribution of low, medium, and high risk by each age group, and the number of patients in each predicted risk group that actually have heart disease. For example, for age group 45-55, 0 out of 2 patients predicted as medium risk had heart disease, while 8 out of 8 patients predicted as high risk had heart disease.

Results and Analysis: The project successfully built a predictive model with 88.3% accuracy. The analysis confirmed well-known clinical risk factors, such as the number of blocked vessels, type of chest pain, and maximum heart rate. The Neural Network performed comparably or better than other models despite the smaller dataset. The risk stratification also shows that the model can be used to predict high-risk subgroups, which should be interpreted with respect to other variables such as age. The statistical tests also support the model output and identify well-known clinical factors such as high blood pressure, maximum heart rate, among others, as highly correlated with heart disease.

Implications: This project illustrates that even with a small dataset, by adapting a neural network to include early stopping criteria and incorporate concepts such as dropout and batch normalization, computational models can align with established clinical understanding of heart disease and serve as a tool for clinical decision-making. This model can be used as a screening assistant to help clinicians prioritize high risk patients, identify patients at risk in the future, and validate the importance of certain metrics with data.

Limitations: Small Dataset: With only 297 samples after cleaning, the model is susceptible to overfitting, and the results may not generalize well to a broader population. The performance of all models should be validated on a larger dataset. **Data Imbalance:** The dataset is skewed towards male patients (68%), which could introduce bias into the model, potentially reducing its accuracy for female patients. According to Cleveland Clinic, female presentation of heart disease and acute effects of heart disease significantly differs from males, which may limit model capabilities. **Simplicity of Model:** The Neural Network architecture is relatively simple to avoid overfitting and adjust for small datasets. The architecture is a simplified and modified version of literature-backed architecture for disease prediction, and this may result in omission of critical architectural components relating to prediction. **Correlation Constraint:** The analysis identifies associations but cannot prove causation between the features and heart disease.

Citation List

1. **Data Source:** Detrano, R. (1989). International application of a new probability algorithm for the diagnosis of coronary artery disease. *American Journal of Cardiology, 64*, 304-310. UCI Machine Learning Repository: Heart Disease Dataset.

2. Neural Network Architectures in Biomedical Applications Srivastava, Prof. (Dr.) N., & Chatterjee, P. (2025). Achieving Superior Predictive Performance through Deep Neural Networks for Enhanced Cardiovascular Disease Detection. International Journal of Research Publication and Reviews, 6(6), 5964–5968. https://doi.org/10.55248/gengpi.6.0625.21113

3. PyTorch Methods, EarlyStop Sourcing, etc.: Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., & Bai, J. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. ArXiv.org. https://arxiv.org/abs/1912.01703